

VOLUME II - EXISTING SOLUTIONS / DOCUMENT 3 OF 3

Gap Analysis and Architectural Recommendation

What is Missing, Why It Is Missing, and the Proposed System Architecture

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ABSTRACT

This document maps the requirements of Volume I Document 3 against the capabilities surveyed in Volume II Documents 1 and 2, identifies the specific gaps, and explains why a mature market has left them open. It then sets out the recommended system architecture - designated the Fog-Independent Haulage Guidance and Awareness System - built on the inversion that the problem is not primarily one of seeing better but of needing to see less. The document closes with a phased implementation roadmap, a risk register and an outline benefit model.

Prepared as part of a solution-design research package addressing dense-fog and low-visibility haul road operations in large-scale mechanised open cast iron ore mines. Figures relating to NMDC production, statutory provisions and commercial systems are compiled from public sources; site-specific values marked as indicative require validation against NMDC operational records before use in a formal proposal.

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1. Requirements-to-Capability Mapping

The table maps each requirement group from Volume I Document 3 to the best available existing capability identified in Volume II Document 1, and records the residual gap.

Requirement group	Best existing capability	Residual gap
FR-01 to FR-04 Visibility classes and nowcasting	Aviation LVP; highway fog systems; forward-scatter meters	No mining implementation. Components mature; integration absent.
FR-05 to FR-10 Positioning and lane guidance	AHS navigation stack; Cat MineStar Terrain	Not offered as a fog-condition guidance product for manned trucks. Primary gap.
FR-11 to FR-17 Non-cooperative hazard detection	Mining CAS radar options; automotive 4D radar	Performance at MOR 3-20 m uncharacterised. Range envelopes of deployed mining PDS far short of requirement. Primary gap.
FR-18 to FR-22 Cooperative awareness	Hexagon CAS, Cat Detect, ADS-B, AIS	Largely closed. Mature and proven at scale.
FR-23 to FR-27 Alerting and intervention	Mining CAS alerting; TCAS logic; ETCS braking curves	Trajectory-aware, visibility-adaptive alerting is partially available; visibility-coupled advisory speed is not offered.
HR-01 to HR-11 Human-machine interface	Mining CAS displays; aviation SVS/CVS; maritime ARPA	Plan-view displays exist. Synthetic perspective guidance view does not exist in mining.
CR-01 to CR-08 Communications	C-V2X, DSRC, proprietary sub-GHz CAS bearers	Largely closed. Bearer selection requires site RF survey.
OR-01 to OR-09 Control room	Mining FMS and CAS analytics platforms	Visibility-overlaid common picture and fog decision support absent.
ER-01 to ER-10 Environmental	Mining-qualified hardware exists across the market	Largely closed; Bailadila-specific ferrous and monsoon validation outstanding.
IR-01 to IR-07 Integration and data	FMS integration is routine	Data governance framework must be created, not procured.
RR-01 to RR-07 Reliability	Mature field practice	Largely closed.
GR-01 to GR-06 Regulatory	DGMS circulars already mandate PDS and CAS	Advisory functions well supported. Intervention and autonomy pathways undeveloped in India.
AR-01 to AR-10 Acceptance	None	No vendor or operator publishes MOR-referenced acceptance evidence. Gap is market-wide.

2. The Identified Gaps

2.1 G1 - Lane-level guidance in zero visibility

Statement. No commercially available mining system tells a manned haul truck operator, in dense fog, where the edge of the road is with sub-metre accuracy.

Mining collision avoidance answers the question "what else is near me?" It does not answer "where is the road?". Yet the road question is the one that determines whether the operator can maintain speed. An operator who knows with confidence that no other vehicle is within 200 m still cannot drive at 25 km/h if they cannot see the berm on a switchback.

The capability exists - autonomous haulage trucks navigate haul roads with precisely this stack - but it is packaged inside a product that removes the driver, costs an order of magnitude more, and carries an undeveloped Indian regulatory pathway.

This is the single most important gap in the survey, and it is the gap whose closure unlocks the largest share of the production benefit.

2.2 G2 - Non-cooperative detection at stopping distance

Statement. Deployed mining proximity detection operates at ranges of roughly 10-15 m, designed for blind-zone and manoeuvring protection. The stopping distance analysis of Volume I Document 3, Section 3.1 requires detection at 50-80 m to support useful haul speeds.

The technology to close this gap is a commodity: automotive 77/79 GHz radar detects vehicle-sized targets to 200-250 m and is essentially unaffected by fog. The gap is not technological but **applicative** - mining proximity products were specified against the pedestrian blind-zone hazard, and the specifications have been inherited by successive product generations.

Additionally, **no surveyed vendor publishes detection performance against calibrated MOR**. A system may in fact perform adequately at MOR 5 m; there is no published evidence either way.

2.3 G3 - Graded instrumented operating modes

Statement. Mining practice responds to fog with a subjective, mine-wide, binary decision. Aviation and highway practice respond with instrumented, zone-specific, graded operating states.

Bailadila fog is strongly stratified by elevation and patchy in the advection regime. A mine-wide declaration necessarily either stops clear sections or permits movement in unclear ones. The instruments to measure MOR per zone are inexpensive and mature. The gap is that no mining product integrates them into an operating-mode framework.

This gap is the cheapest to close and addresses levers 3 and 4 of the loss model - the number of stop-start events and the restart overhead - which, as established in Volume I Document 1, Section 4.2, are routinely underestimated.

2.4 G4 - Integrated common operating picture

Statement. Fleet management shows dispatch and productivity. Collision avoidance analytics shows safety events. Neither shows visibility. None combines vehicles, visibility, road state and production consequence into one picture from which a control room can make a graded fog decision.

The consequence is that the decision to suspend haulage is made without a quantitative view of its cost, which biases it toward conservatism - correctly, given the information available, but at a production price that no one is currently measuring.

2.5 G5 - Characterised dense-fog performance

Statement. The industry has no published, MOR-referenced performance data for any collision avoidance or proximity system.

This is a market-wide evidence gap with two consequences. First, NMDC cannot procure on evidence; it must procure on specification and then validate. Second - and more usefully - **a system that is characterised against measured MOR would be the first**, and that characterisation is inexpensive relative to the programme cost. Requirement AR-02 exists precisely to close this gap, and the resulting dataset would have value beyond Bailadila.

2.6 G6 - Mixed-fleet economics

Statement. Capable systems are either OEM-native (limiting mixed-fleet coverage) or autonomy-class (economically inapplicable to a brownfield retrofit). The gap is a capable, vendor-neutral, retrofittable system at a per-vehicle cost appropriate to a large Indian mixed fleet.

Partial fleet coverage is not merely a commercial issue but a safety one: an uninstrumented machine is invisible to the cooperative layer, and an operator who has learned to trust the display may not account for it. This is failure mode F4 of Volume I Document 2.

3. Why the Gaps Exist

Understanding the cause matters, because it indicates whether a gap is likely to be closed by the market before NMDC needs it.

- ? **Products were specified against a different hazard.** The dominant mining fatality mechanism addressed by regulation has been the pedestrian-machine blind-zone interaction. Products were correctly specified against it. The long-range, high-closing-rate, fog-driven haul road problem was never the design case.
- ? **Fog is not the dominant visibility problem in the reference markets.** The global open-pit technology market is shaped by the Pilbara, Chile, Nevada, Mongolia and Africa. The visibility problem in those markets is **dust**, which is optically severe but behaves differently from water fog and is often self-generated and therefore controllable by watering and by spacing. A monsoon orographic cloud sitting on a working bench for days is a niche condition in global market terms, and a defining condition at Bailadila.
- ? **Demonstrations happen in clear weather.** Commercial incentives favour capabilities that are visible in a demonstration. Camera-based detection demonstrates beautifully. Radar performance at MOR 5 m does not demonstrate at all unless the buyer waits for fog and brings an instrument.
- ? **Autonomy absorbed the investment.** The industry's positioning and guidance investment went into removing the driver rather than into assisting one. Driver assistance for manned haul trucks has been comparatively neglected, despite serving a far larger installed fleet at far lower cost.
- ? **Regulation followed the same path.** DGMS circulars mandate PDS and rear-view cameras - the blind-zone controls - and the market responded to the mandate. There is no mandate for haul road guidance in fog, so there is no product.

The relevant conclusion for NMDC: **these gaps will not close on their own on the timescale of the 2030 ramp.** The market has no strong incentive to close them, and the constituent technologies are already commoditised elsewhere. The gap is an integration and specification gap, which means it can be closed by a determined operator working with an integrator - and that the resulting capability is defensible and transferable to other Indian monsoon-affected mines.

4. The Architectural Inversion

The problem as commonly framed is: *how do we let the operator see through fog?*

Framed that way, it is close to unsolvable at MOR 3-5 m. Thermal penetrates partially. LiDAR fails. Image enhancement cannot recover a signal that never arrived. Every approach is a losing fight against Mie scattering.

The aviation precedent suggests a different framing: *how do we reduce what the operator needs to see?*

That question has a complete answer, and aviation has been executing it since the 1930s.

- ? The **static world** - where the road is, where it curves, where it descends, where the berm is, where the junction is, where the tipping edge is - **does not need to be seen at all**. It is known in advance. It only needs to be *located*, which requires precise positioning against an accurate map. This is Synthetic Vision.
- ? The **cooperative world** - every other truck, dozer, grader, water tanker, light vehicle and equipped person - **does not need to be seen either**. Each one knows where it is and can say so over a radio link that fog does not touch. This is ADS-B, AIS and mining CAS.
- ? Only the **residual world** - rockfall, a washed-out edge, a stalled unequipped vehicle, a person without a tag, an animal - genuinely requires sensing. This is a much smaller problem than "see everything", and it is exactly the problem that a 77/79 GHz radar solves well.

The inversion: the static world comes from the map, the cooperative world comes from the radio, and only the residual world needs a sensor. Framed this way, dense fog reduces a complete perception problem to a bounded detection problem - and the bounded problem has a commodity solution.

Two corollaries follow, both important.

First, the map is safety-critical. If the system draws the road from a digital model and the model is stale - a ramp extended, a berm moved, a section closed - the display will confidently show a road that is not there. This elevates the digital road model from a convenience to a safety-critical asset requiring version control, validity timestamps, field update capability within a shift (FR-07) and visible staleness indication.

Second, the residual sensor is the last line of defence and must never be omitted. The temptation with a map-and-radio architecture is to economise on the radar layer, since most of the picture comes free. That would be an error: the radar layer covers exactly the hazards that the map and radio cannot know about, and those hazards - washouts, rockfalls, stalled vehicles - become *more* common during the monsoon, not less.

5. Recommended Architecture - FIHGAS

Fog-Independent Haulage Guidance and Awareness System.

5.1 Design principles

- ? **P1 - Fog-independence by physics, not by processing.** Every primary layer operates outside the Mie scattering peak for water droplets. Optical layers contribute only where they are valid.
- ? **P2 - Know rather than see.** The static world comes from the map, the cooperative world from the radio.
- ? **P3 - Two independent primary layers.** Cooperative RF and radar are independent in failure mode. Neither single failure blinds the system.
- ? **P4 - On-vehicle autonomy.** All safety processing on the vehicle. Infrastructure improves coordination, never protection.
- ? **P5 - Graded, instrumented response.** Operating mode from measured MOR per zone, not from subjective judgement.
- ? **P6 - Honest degradation.** The system always shows the operator what it can and cannot currently see. Silent partial failure is designed out.
- ? **P7 - Advisory first, intervention later, architecturally separated.** Deployment of advisory function is never gated by intervention approval.
- ? **P8 - Retrofit and vendor-neutral.** Open interfaces, commodity components, documented harnesses, fleet-wide coverage.

5.2 Layered architecture

Layer	Contents	Location
L0 - Environment	Forward-scatter visibility meters by elevation zone; reference transmissometer; weather sensors; nowcasting engine	Roadside, solar-powered; control room compute
L1 - Position	RTK GNSS dual-antenna, tactical IMU, wheel odometry, radar odometry aiding; integrity monitor	Vehicle; RTK bases fixed
L2 - Map	Digital haul road model: centreline, edges, lanes, gradient, curvature, speed zones, junctions, tipping edges, exclusion zones; version and validity	Vehicle cache; control room master
L3 - Cooperative	V2V broadcast at 10 Hz of identity, class, position, confidence, velocity, heading, state; personal devices; hazard markers	Vehicle; personal devices
L4 - Residual sensing	4D imaging radar forward and quarters; short-range radar 360 deg; magnetic-field PDS for personnel; LWIR thermal and cameras as secondary	Vehicle
L5 - Fusion and threat	Track fusion across L3 and L4; association and duplicate resolution; trajectory prediction against L2; time-to-collision and time-to-lane-departure; graded alert decision; self-degradation detection	Vehicle compute
L6 - Interface	Plan view primary; synthetic perspective view promoted at V3-V4; directional audio; haptic; persistent health, mode and effective range indication	Cab
L7 - Coordination	Common operating picture; visibility overlay; movement authority and slot management; re-dispatch; event log; fleet health; FMS integration	Control room
L8 - Analytics	Nowcast model training; KPI reporting; near-miss analysis; alert tuning; optional digital twin	Off-line

5.3 Functional flow

- ? L0 measures MOR per zone and publishes the current visibility class, with a 15-30 minute nowcast.
- ? L7 sets the operating class for each zone and, at V3-V4, issues movement authority or slot allocation on critical sections.
- ? On each vehicle, L1 produces a continuous, integrity-monitored position and heading; L2 supplies the road geometry around that position.
- ? L6 draws the road corridor and the host vehicle's lateral position within its lane - **this alone restores the operator's primary spatial reference regardless of visibility.**
- ? L3 populates the picture with every equipped vehicle and person, with identity and intent.
- ? L4 sweeps the corridor for anything L2 and L3 do not account for.
- ? L5 fuses, predicts against the road geometry, and issues graded alerts on time-to-conflict, adapting thresholds to class, speed, load and gradient.
- ? L5 continuously assesses its own capability and, on fouling, fault or positional confidence loss, declares a degraded mode that L6 makes unmistakable and L7 records.
- ? L7 maintains the common picture and the event log; L8 closes the loop on nowcast skill, alert tuning and KPI reporting.

5.4 Degradation behaviour

Designed degradation is what separates a system that is trusted from one that is disabled.

Failure	System behaviour	Operating consequence
GNSS outage within INS coasting limit	Position continues on INS and odometry; confidence indicator falls	Lane guidance continues with widened lane margin displayed
GNSS outage beyond coasting limit	Lane guidance withdrawn and visibly so; cooperative and radar layers continue	Revert to V4 procedures; collision protection intact
Radar fouled or failed	Declared degraded; effective range indication drops to zero; cooperative layer continues	Non-cooperative protection lost; operating class restricted by L7
Cooperative link lost	Radar and map layers continue; display marks cooperative layer unavailable	Reduced awareness of equipped traffic; radar still detects them as non-cooperative
Road model stale beyond tolerance	Synthetic road drawn in a distinct degraded style with explicit staleness warning	Operator instructed not to rely on lane guidance in affected section
Backhaul lost	All vehicle functions continue unaffected; control room picture degrades; events buffered	Coordination degraded, protection unaffected
Vehicle unit total failure	Heartbeat lost at control room; vehicle flagged; other vehicles see it only on radar	Vehicle withdrawn or restricted per procedure

The principle behind this table: **the system never presents a normal-looking picture that it cannot support.** Every degradation is visible to the operator and to the control room. This directly addresses failure modes F3 (silent failure) and F5 (over-trust), which between them account for most of the ways a technically sound system becomes dangerous.

6. Gap Closure Mapping

Gap	Closed by	Residual risk
G1 Lane-level guidance in zero visibility	L1 + L2 + L6 synthetic perspective view	Digital road model currency becomes safety-critical; mitigated by FR-07 and staleness indication
G2 Non-cooperative detection at stopping distance	L4 4D imaging radar at 77/79 GHz	Performance at MOR 3-5 m must be validated on site per AR-02/AR-03
G3 Graded instrumented operating modes	L0 + L7	Requires procedural change and DGMS engagement per GR-05
G4 Integrated common operating picture	L7 with FMS integration	Integration effort with existing systems; mitigated by IR-01, IR-02
G5 Characterised dense-fog performance	AR-02 instrumented acceptance regime	Requires monsoon-scheduled testing; programme timeline risk
G6 Mixed-fleet economics	Commodity automotive components, vendor-neutral open interfaces, documented retrofit kits	Fleet coverage must be complete enough to avoid F4

7. Phased Implementation Roadmap

Phase	Scope	Duration (indicative)	Value delivered
Phase 0 - Baseline	Deploy visibility sensor network by elevation zone. Instrument the loss: extract fog stoppage hours, stop-start counts, fog-condition cycle times from FMS and shift records for prior seasons. Establish KPI baselines. Begin site RF propagation survey. Begin DGMS engagement.	3-4 months, must span a monsoon	The business case, the acceptance baseline, and the nowcast training data. Lowest cost item in the programme and prerequisite for everything else.
Phase 1 - Awareness	Cooperative V2V across the full fleet plus personal devices. RTK base infrastructure. Basic in-cab plan view. Control room common picture with visibility overlay. FMS integration.	6-9 months	Fleet-wide cooperative awareness independent of visibility. V2 operation becomes defensible. Satisfies and exceeds existing DGMS PDS mandates.
Phase 2 - Guidance	GNSS/INS on all haul trucks. Digital road model creation and field update workflow. Synthetic perspective guidance view. 4D imaging radar forward and quarters. Full fusion and graded alerting. Visibility-coupled advisory speed.	9-15 months	The core capability. V3 guided operation becomes defensible. This is where the majority of the production recovery is realised.
Phase 3 - Optimisation	Nowcasting engine. Slot management and movement authority on critical sections. Thermal secondary layer. Vehicle-derived visibility sensing. Alert tuning against measured false-alert data. Digital twin.	9-12 months	Reduced stop-start count and restart overhead. Anticipatory rather than reactive fleet management.
Phase 4 - Intervention	Advisory speed enforcement or automatic braking, subject to DGMS approval. Optional selective autonomy on fixed routes.	18+ months, approval-gated	Residual risk reduction. Deliberately last; must never gate Phases 1-3.

Phase 0 is the phase most likely to be skipped and the one least affordable to skip. Without baseline measurement the programme cannot demonstrate its benefit, cannot size its business case, cannot train its nowcast model, and cannot conduct the MOR-referenced acceptance testing of AR-02. It also costs least. It should begin before procurement of anything else.

8. Risk Register

ID	Risk	Likelihood	Impact	Mitigation
R1	Radar performance at MOR 3-5 m proves inadequate on site	Low	Critical	Early instrumented trial in Phase 0/1; fall back to cooperative-only with reduced operating classes; evaluate 79 GHz and low-THz options
R2	GNSS availability insufficient in deep pit and switchback geometry	Medium	High	Tactical-grade INS sizing; radar odometry aiding; augment with local RF trilateration on worst sections; survey GNSS availability in Phase 0
R3	Nuisance alert rate exceeds HR-06; operators disengage	High	Critical	Trajectory-aware alerting from the outset; measure from week one; single-action nuisance reporting; dedicated tuning resource; treat as a headline KPI

ID	Risk	Likelihood	Impact	Mitigation
R4	Digital road model goes stale; system shows a road that is not there	Medium	Critical	Field update within a shift; validity timestamp; automatic staleness detection by comparing driven paths against the model; distinct degraded rendering
R5	Operator over-trust leads to excessive speed in fog	Medium	Critical	Persistent effective-range display; visibility-coupled advisory speed; explicit training on limits; monitor speed distribution against visibility class
R6	Industrial relations resistance to perceived surveillance	Medium	High	Data governance agreed before deployment per IR-04; introduce as a safety tool; operator involvement in interface design and alert tuning
R7	Partial fleet fitment creates false confidence	High	High	Radar layer mandatory so unequipped machines remain detectable; explicit cooperative-versus-sensed indication; prioritise 100% fitment of the highest-risk classes
R8	DGMS approval delay for intervention functions	Medium	Medium	Architectural separation per P7; engage DGMS from Phase 0; Phases 1-3 require no intervention approval
R9	Spectrum authorisation delay for chosen V2V bearer	Medium	High	Confirm regulatory position in Phase 0; carry a licence-exempt fallback bearer in the evaluation
R10	Monsoon-scheduled acceptance testing slips a full year	Medium	High	Plan the programme calendar backwards from monsoon windows; treat monsoon availability as a hard schedule constraint
R11	Sensor fouling degrades performance faster than anticipated	Medium	High	Radar-first architecture is inherently fouling-tolerant; self-detection per FR-17; measure fouling rates in Phase 0 with instrumented test units
R12	Integration with existing FMS proves harder than expected	Medium	Medium	Establish interface feasibility in Phase 0; require open interfaces per IR-01 in all procurement
R13	Security situation disrupts deployment or remote infrastructure	Medium	Medium	Minimise unattended remote assets; tamper reporting per IR-06; phase deployment by area
R14	Per-vehicle cost escalates beyond fleet economics	Medium	High	Commodity automotive components; vendor-neutral specification; competitive procurement; phase to allow cost learning

R3 and R7 are rated the highest combined risk in this register, and both are human-factors rather than technical risks. This is consistent with the sector-wide record set out in Volume I Document 2, Section 8: systems of this class fail operationally far more often than they fail technically.

9. Outline Benefit Model

The model is presented as a structure for NMDC to populate from its own records. No value is asserted here that is not derivable from site data.

Inputs required from NMDC.

? Monsoon-month production and despatch, by complex, for at least three prior years

- ? Scheduled versus effective haulage hours during monsoon months
- ? Recorded visibility-related stoppages: count, duration, and cause where logged
- ? Fog-condition versus dry-condition mean cycle times, from FMS
- ? Fleet size, class mix and mean productivity per effective fleet hour
- ? Realised value per tonne and marginal contribution per tonne
- ? Monsoon-period incident and near-miss record

Benefit structure.

Benefit stream	Mechanism	Corresponding lever (Vol. I Doc. 1 Sec. 4.3)
B1 Recovered stoppage hours	V2/V3 become productive classes rather than suspension triggers	f_stop
B2 Higher retained speed	Guidance restores confidence to maintain speed in fog	d_speed
B3 Fewer stop-start events	Nowcasting and graded modes replace reactive binary stoppage	n_restart
B4 Faster restart	Control room re-sequencing and automated re-dispatch	t_r
B5 Avoided incident cost	Reduced collision and run-over exposure: direct damage, downtime, investigation, statutory consequence	n/a
B6 Improved fleet utilisation	Reduced shovel hang time as truck arrival rate is sustained	Indirect
B7 Reduced schedule pressure	Less dry-season catch-up load; smoother plant and rail loading	Indirect
B8 Enabling the ramp	Removes a binding constraint on Bailadila's path to ~80 MTPA	Strategic

Cost structure. Per-vehicle hardware and installation across the fleet; RTK and visibility infrastructure; control room integration; digital road model creation and ongoing maintenance; training; annual maintenance, calibration and spares; programme management and DGMS engagement.

The strategic argument (B8) is likely to dominate the arithmetic. At 37 MTPA a monsoon productivity shortfall is recoverable in the dry season. At 80 MTPA, with the fleet and haul road network operating far closer to their geometric limits and with no spare dry-season capacity to absorb catch-up, it is not. The system should therefore be evaluated not only on the tonnes it recovers today but on whether the 2030 target is achievable without it.

10. Recommendations

- ? **Adopt the architectural inversion.** Specify and procure against "know rather than see" - precise positioning against a maintained digital road model, cooperative broadcast, and radar for the residual - rather than against improved visibility. Reject any proposal whose primary forward sensing layer is optical.
- ? **Begin Phase 0 immediately and independently of any procurement decision.** Deploy visibility sensors, extract the historical loss baseline from existing FMS and shift records, survey GNSS availability and RF propagation on site, and open the DGMS dialogue. Phase 0 is inexpensive, has no dependencies, and everything downstream depends on it.

- ? **Make MOR-referenced acceptance testing a non-negotiable contractual requirement.** Requirement AR-02 should appear in any tender. Without it, NMDC will procure clear-weather performance for a fog problem.
- ? **Treat the nuisance alert rate (HR-06) as a headline acceptance criterion,** measured from the first week of trial, with a named owner and a tuning budget. It is the highest-likelihood, highest-impact risk in the register.
- ? **Require complete fleet coverage of the cooperative layer before relying on it,** and retain the radar layer as mandatory precisely so that partial coverage does not create false confidence.
- ? **Separate advisory from intervention architecturally and contractually,** so that the majority of the benefit is realised on a timeline that is not gated by the approval of control-authority functions.
- ? **Agree the data governance framework before any vehicle is fitted.** This is deployment-critical, cannot be retrofitted after trust is lost, and costs nothing but attention.
- ? **Plan the programme calendar backwards from monsoon windows.** Fog-condition validation can only occur in a limited annual window; a schedule that misses it loses a full year.
- ? **Treat the resulting MOR-referenced performance dataset as an asset.** It would be the first of its kind publicly available in mining, is directly transferable to other Indian monsoon-affected operations, and materially strengthens NMDC's position in specifying and procuring future systems.